MAD PROJECT

[Reduce Knox’s Plastic Use]

CLASS: ENVS101-1

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01

ABOUT

single-used plastic

single-use plastics have horrific effects on the environment

Throughout their life cycle, plastics wreak havoc on the ecosystems they come in contact with

Stage. 1

production

emit toxins which are absorbed by the atmosphere + soil

harm human health as well as the atmospheric ability to regulate weather patterns and soil health

Stage. 2

usage

are sent either to be recycled / to landfills

much of this plastic finds its way into oceanic and terrestrial ecosystems

Stage. 3

build-up

cause harm to animal + plant life

Stage. 4

breakdown

create microplastics that end up in our food + water systems

The production of plastic is unfavorable for us all, therefore any reductions we can make in our consumption are essential to a healthy future.

02

Problem

1-Concern

single-use plastics

WHY?
WHY?
WHY?

Plastic Concern

observation	C-Store / week
750	plastic bags
600	plastic water bottles

Knox is a consumer of many single-use plastics

Can we help reduce Knox's plastic footprint?

03

Solution

2-IDEAS

alternative “plastic”

Replace plastic shopping bags and plastic water bottles with:

compostable shopping bags
+
boxed water

water bottle filler

installation of at minimum one water bottle filler per stairwell in <u>each residence hall</u>

[6 total]

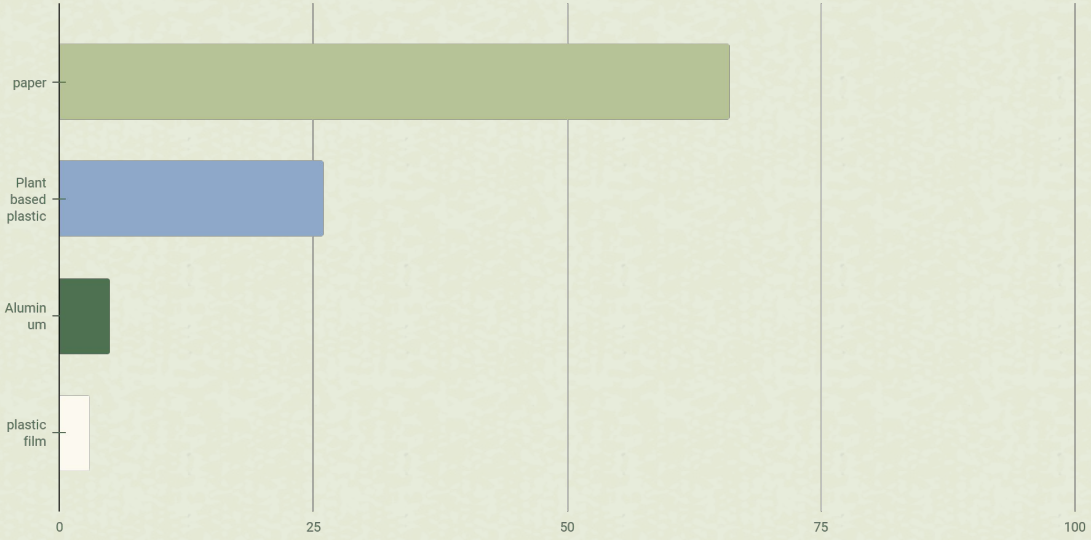
Why compostable shopping bags ?

Takes **10-60** days

to degrade in the soil
[depending on favorable environment conditions]

To compare, traditional plastic bags take hundreds of years to fully decompose

Why boxed water bottles_ ?



Paper
66%

Plant-based plastic
26%

Aluminum
5%

Plastic film
3%

consists of 92% plant-based content = eco-friendly

Why WaterBottleFiller.exe ?

provide [clean+accessible] water to all students

students would be less inclined to buy packaged water

Knox's commitment to working towards a cleaner future

minimum:

3000 gals ≅ **19,200** 20oz plastic bottles

04

Compare

Learn from ALDI supermaket system

Paper bag pay extra
Tote bag pay extra
Your bag free **?**

reducing and recycling plastic

= smaller plastic footprint
= more eco-friendly

Apply to Gizmo+C-Store?

05

TheCons

the overall cost might be a disadvantage of these alternatives

It is all about the price!!!

about $1,000 / water bottle filler

[price] 5x Plastic water bottles = boxed water bottles

[price] 2x Plastic bag = compostable bags

Thanks!

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SOURCES: IPCC AR6; NASA

PAUL HORN / Inside Climate News

CF4, the primary PFC released by Century Aluminum, is 7,380 times worse for climate change than carbon dioxide on a ton-for-ton basis over a 100-year period. But, unlike CO2, which remains in the atmosphere for approximately 300-1,000 years, CF4 remains in the atmosphere for 50,000 years.

In 2021, the Sebree plant, the largest U.S. aluminum production facility